

Essential oils in plants: Plant physiology, the chemical composition of the oil, and natural variation of the oils (chemotaxonomy and environmental effects, etc.)

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1.1 Introduction

Essential oils are extricated from different bioactive rich plants exhibiting aroma. These essential oils (EOs) are natural, volatile complex constituents characterized by the aroma of their respective plants (Mironescu and Georgescu, 2021). A plethora of applications have been recorded in the food, cosmetics, perfumeries, and pharmaceutical industries owing to their antioxidant and antimicrobial activities (Dhifi et al., 2016; Mironescu and Georgescu, 2021). As per the “International Standard Organization on Essential oils (ISO 9235:2013)” and “European Pharmacopoeia (Council of Europe 2004),” EOs are defined as the output retained by extracting through hydrodistillation, steam distillation, or dry distillation or via another suitable mechanical extraction process, used for Citrus fruits only (cold pressing generally used as the bioactive constituents are thermosensitive and unstable). As per the definition,

other aromatic products extracted using solvent-based extrication, supercritical fluid extraction (SFE), and microwave-assisted extraction (MAE) are excluded (Hanif et al., 2019).

Worldwide production of EOs is evaluated to be greater than 150,000 tons with revenue generation of USD 6B, approximately increasing three times than 1990 with a 50% increase observed since 2007. Different economic analyses have indicated the amount to reach 370,000 tons annually after the 2020s with a value of more than USD 10B (Barbieri and Borsotto, 2018). The major producers of EOs include China, India, Indonesia, Sri Lanka, and Vietnam while minor contributions are also obtained from African countries including Morocco, Tunisia, Uganda and Ethiopia, Egypt, and Algeria; the Ivory Coast, South Africa, Ghana, Kenya, and Tanzania (Barbieri and Borsotto, 2018). This increase in consumption is owing to the demand from these principal markets: food and beverage (35%), fragrances, aromatherapy and cosmetics (29%), household (16%), and pharmaceutical (15%). The food and beverage industry mainly employs essential oils from orange, lemon, and lime (Manion and Widde, 2017).

EOs are also distinguished with different names including essences, volatile oils, etheric oils, or aetheroleum. They contain more than 300 different compounds containing hydrocarbons (in natural mixtures) (terpenes), oxygen- (alcohols, aldehydes, ketones, carboxylic acids, esters, lactones), and sulfur-containing (sulfides, disulfides, trisulfides) organic constituents (Conde-Hernández et al., 2017). EOs are obtained from different plant parts including roots, shoots, flowers, fruits, and seeds; where they are initially produced and stored in the secretory structures. These secretory structures vary in their morphological, functional, and distribution sections as they reduce the autotoxicity. Being found on either the internal and external plant organs surface or within the tissues structure, they are considered as external or internal secretory structures, respectively. Internal structures consist of idioblasts, secretory cavities, and ducts both while the counterparts contain glandular trichomes, epidermal cells, and osmophores. Some storage organs in roots, tubers, and wood sections require mechanical degradation to facilitate extraction from the oil storage organs (De Sousa, 2015).

The compositional profile varies according to both intrinsic and extrinsic factors. Intrinsic include genetically variations, sexually, seasonal and ontogenetic based while extrinsic ones include both environment and ecological-based (Mironescu and Georgescu, 2021). Among different sources, not only does the chemical composition varies but also the bioactive constituents obtained contribute to numerous functional contributions. These EOs have antimicrobial, antioxidant, anticancer, acaridical, herbal, and therapeutic contributions. For instance, cinnamon EO exhibited the highest antibacterial effect while the fumigant potential was recorded in both cinnamon, clove, and anise EOs (Tu et al., 2018). This chapter will highlight the phytochemistry, chemical characteristics, and administration of different essential oils obtained from the plants while considering the safety concerns related to them in different industrial applications.

1.2 Phytochemistry of different essential oils

Phytochemicals are nonnutrimental chemical constituents or secondary metabolites generated by plants through many chemical pathways (Budisan et al., 2017). They exhibit a critical role in physiological activities of plant growth such as symbiotic relationships,

reproduction, and their interactions with the environment and other organisms. Phytochemicals include flavonoids, polyphenols, steroidal saponins, vitamins, and organosulfur compounds (Forni et al., 2017). The majority of phytochemicals, food, herbal items, and beverages are frequently referred to as “nutraceuticals” in the literature, focusing on their health-promoting effects, including the prevention as well as treatment of cardiovascular disease, cancers, Alzheimer’s disease along with some neural disorders (Winter et al., 2017).

1.2.1 Major phytochemicals in the plant-based essential oils

Essential oils are made up of a complex blend of chemicals at different concentrations ranging from 20 to 60 (Chouhan et al., 2017). Some major phytochemicals in plant-based essential oils are given below:

1.2.1.1 Terpenes

Terpenes, commonly known as terpenoids, are the largest and most diversified family of naturally occurring chemicals. They are divided as mono, di, tri, tetra, and sesquiterpenes based on the number of isoprene units they contain. They’re mostly found in plants and comprise the majority of essential oils. Terpenes have an important and diverse role among the natural substances that deliver medical benefits to an organism. Terpenes can be found in thyme, cannabis, tea, Spanish sage, and citrus fruits (e.g., lemon, orange, and mandarin). Terpenes offer a wide range of therapeutic potential, including antiparasmodial activity, which is significant because its mechanism of action is comparable to that of the widely used antimalarial drug chloroquine. Antiviral effects of monoterpenes, in particular, have been intensively studied. Terpenes have the potential to operate as antidiabetic and anticancer agents, which is highly crucial due to the rise in the incidence of cancer and diabetes. Besides these properties, Terpenes also can suppress the side effects and change the route of administration (Cox-Georgian et al., 2019). They have a complex structure with a wide range of effects and different mechanisms of action (Yang et al., 2020). The chemical structures of classes of terpenes are provided in Fig. 1.1 below:

1.2.1.2 Alpha-pinene

Alpha-pinene is among two isomers of pinene, a terpene chemical component. It can be found in the oils of a variety of coniferous plants, the most prominent of which being the pine. This molecule is usually found in essential oils and is used in oils and diffusers as a flavoring ingredient. The racemic combination is found in several oils, such as eucalyptus oil and orange peel oil. Essential oils’ antiinflammatory properties are linked to this molecule, which has a powerful antiinflammatory effect (Özbek and Yılmaz, 2017). The chemical structure of the Alpha-Pinene is presented in Fig. 1.2.

1.2.1.3 Steroids

Steroids are a class of phytochemicals produced by plants. Sterol and its derivatives are examples of steroids. The chemical structure of steroids is provided below in Fig. 1.2. The three primary types of sterols are zymosterol, zoosterol, and phytosterol. The largest concentrations of steroids are found in foods including vegetable oils, nuts, and seeds. They are also

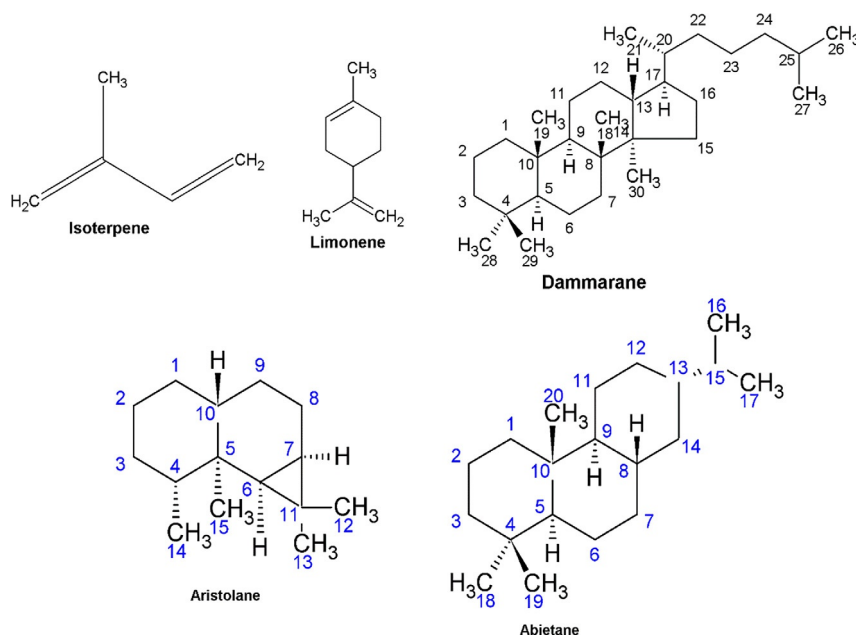


FIG. 1.1 Chemical structures of different classes of terpenes including isoterpene unit, moniterpene (limonene), sesquiterpenoid (aristolane), diterpene (abietane), and tetracyclic triterpene (dammarane).

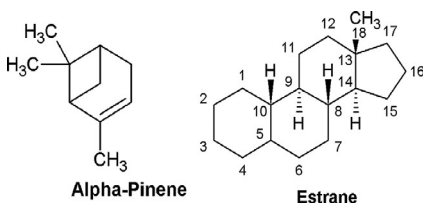


FIG. 1.2 Chemical structures of alpha-pinene and steroid (estrange).

found in plant-based essential oils like citrus and bergamot oils as essential oils are rich in essence-giving chemicals. Steroids provide certain benefits to plants as well as to humans after consumption. Plants utilize steroids in their metabolism. For humans, they are a kind of cholesterol-like molecule and the most widely utilized cholesterol-lowering agent by restricting the amount of cholesterol that may enter the body (Siano and Cautela, 2020). Moreover, steroids are utilized in medicine for the management of some diseases. The most frequent sterol derivations are bile acid and vitamin D, both of which are crucial for growth and development (Kolekar et al., 2019).

1.2.1.4 Tannins

Tannin, commonly recognized as tannic acid, is a collection of phenolic chemicals found in woody flowering plants as well as essential oils. Tannins are stashed in vacuoles within the plant cell as bioactive compounds, which shield the other cell structures. Many plants have

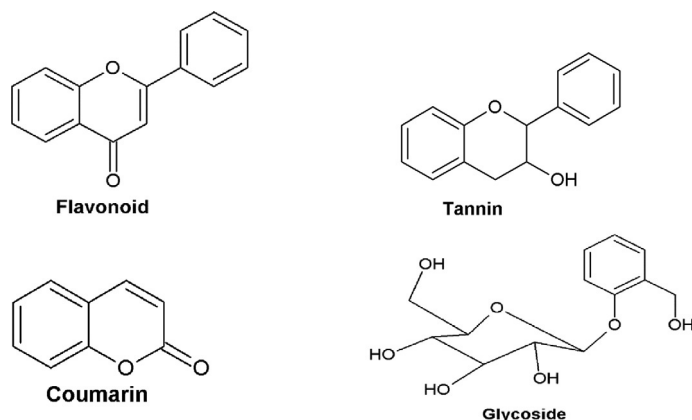


FIG. 1.3 Chemical structures of flavonoid, tannin, coumarin, and glycoside.

them naturally in their roots, wood, bark, leaves, and fruit, especially the bark of oak species, sumac, and myrobalan species. They can also be found in galls, which are diseased lesions caused by insect bites ([Jafari et al., 2019](#)). Tannins act as herbivore repellents and have a variety of commercial uses. They are nonvolatile chemicals generated in leaves that are toxic and are frequently employed by plants to ward off predatory animals. Tannins' intrinsic antioxidant and antibacterial action are attributed to their phenolic constituents. Their antioxidant action is determined by their structure and redox characteristics, which are crucial in the absorption and neutralization of free radicals as well as the breakdown of peroxides. Tannins have been demonstrated to have positive qualities to help protect the environment ([Cano et al., 2020](#)). The chemical structure of Tannin is given in [Fig. 1.3](#).

1.2.1.5 Flavonoids

Flavonoids are chemically complex secondary metabolites that serve a range of roles in plants. They are water-soluble and are polyphenolic compounds with 15 carbon atoms. Flavonoids are the most prevalent form of polyphenolic chemical present in essential oils. Some flavonoids are found in high concentrations in plants' oil glands, whereas others were found in areas where carotenoids were present. Certain flavonoids markers detected are present in essential oil-rich locations and carotenoids. Some flavonoids also block specific spores, which helps to protect plants from illness ([Kaurinovic and Vastag, 2019](#)). They provide a wide range of pharmacological and physiological benefits for humans, including free radical scavenging ability, antioxidative activity, antiatherosclerotic, hepatoprotective, prevention of coronary heart disease, anticancer, and antiinflammatory properties ([Górniak et al., 2019](#)). The chemical structure of Flavonoid is given in [Fig. 1.3](#).

1.2.1.6 Glycosides

Glycosides are generally plant-derived chemicals mostly found in essential oils. One or more sugars are coupled with alcohol, phenol, or a complex molecule such as a steroid nucleus to form these compounds. Because they do not contain nitrogen, they are incorrectly referred to be alkaloids. Glycosides are used as medicines and have various pharmacological

properties. One of the important uses of glycosides is treating cardiac failure and irregular heartbeats and they are specifically known as cardiac glycosides (Saroj et al., 2020). Cardiac glycosides may be found in a wide variety of plants, including *Digitalis purpurea* and *Digitalis lanata*. Cardiac glycosides work by increasing calcium availability to cardiac fibers and blocking sodium-potassium-adenosine triphosphatase, resulting in higher intracellular sodium and lower intracellular potassium levels (Hosseini et al., 2019). The chemical structure of Glycosides is given in Fig. 1.3.

1.2.1.7 Alkaloids

Alkaloids are tiny secondary metabolites of plants that contain nitrogen in a ring; alkaloids make up around 20% of plant species. These are natural chemicals that prevent herbivorous animals due to their bitter taste. Most of the plant essential oils are rich in alkaloids. They are utilized as natural insecticides in some plants as alkaloids shield against the harmful activities of some insect species. Plant alkaloids include morphine, which is used to relieve muscle spasms, emetine, and cephalin. They are used as cures for intoxication; caffeine, which has stimulant characteristics; and quinine, which has antimalarial effects and a slightly bitter taste (Fialova et al., 2017). The chemical structures of Alkaloids are given in Fig. 1.4.

1.2.1.8 Phenols

Phenolic compounds are undoubtedly the most extensively studied natural compounds due to their potential health benefits. Generic terms “phenolics,” “polyphenolics,” or “phenolic compounds” refer to more than 8000 compounds found in the plant kingdom. A phenolic compound’s fundamental structure consists of an aromatic ring with one or more hydroxyl (–OH) groups including functional derivatives like esters glycosides and methyl ethers. Naturally, phenolics contain diverse structures from simple to complex. As a result, the presence of –OH groups in the structure of phenolics can be linked to their antibacterial properties. The amount and position of double bonds in the structure influence the relative toxicity of phenolics against microorganisms (Bhuyan and Basu, 2017). Phenolics are usually concentrated in essential oils and give them aromatic properties. Most citrus essential oils contain phenols and have a specific aroma (Ahmed et al., 2019). The chemical structure of Phenol is given in Fig. 1.5.

1.2.1.9 Coumarin

Coumarins are a group of secondary metabolites obtained from plants that are formed through the phenylpropanoid pathway. Coumarins are present in a wide variety of plant

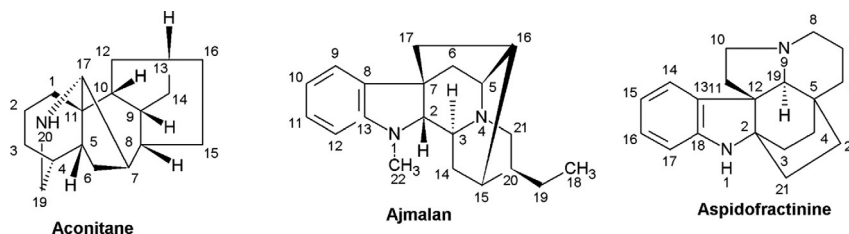


FIG. 1.4 Chemical structures of alkaloids.

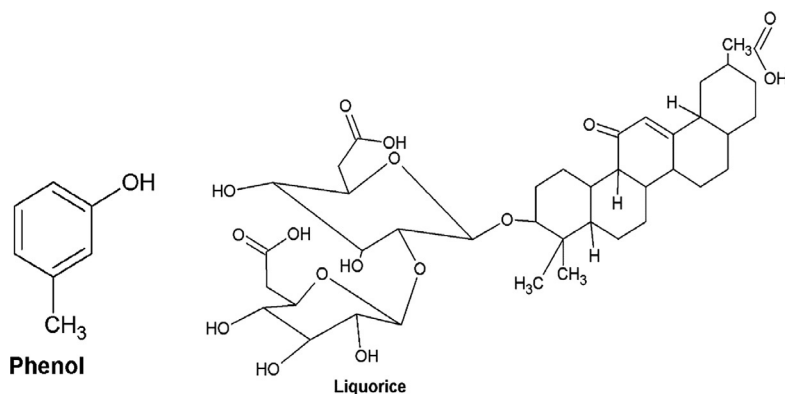


FIG. 1.5 Chemical structures of phenol and saponin (liquorice).

species including essential oils of Rubiaceae, and Poaceae plants, and are best recognized in the form of Citrus, Bergamini's, and bergamot oil. They have a role in plant defense against phytopathogens, environmental stresses response, oxidative stress control, and perhaps as signaling molecules. Coumarins have been widely investigated for their capacity to fight infections in both plants and animals, in addition to their function in iron absorption (Silveira et al., 2020). The chemical structure of Coumarin is given in Fig. 1.5.

1.2.1.10 Saponins

Saponins are derived from the roots of the soapwort plant (Saponaria), which were once used as soap. They are usually found in onion and garlic essential oils. Saponins decrease blood lipids, reduce cancer risks, and improve blood glucose control. A high-saponin diet can help prevent dental cavities and platelet clumping, as well as cure hypercalciuria in humans, and act as an antidote to acute lead poisoning (Runde, 2020). A list of major phytochemicals in different essential oils is given in Table 1.1. The chemical structure of Saponin is given in Fig. 1.5.

TABLE 1.1 Phytochemicals present in different plant-based essential oils.

Phytochemicals	Essential oils	Citation
Terpenes	Tea, thyme, cannabis, Spanish sage, and citrus fruits (e.g., lemon, orange, mandarin)	Cox-Georgian et al. (2019)
Alpha-pinene	Coniferous plants (pines), rosemary, eucalyptus, orange peel oil, cypress, juniper berry, frankincense, helichrysum, wild orange	Kumar et al. (2021); Özbek and Yilmaz (2017)
Steroids	Citrus, bergamot	Siano and Cautela (2020)
Tannins	Eucalyptus, peppermint, fennel, rosemary, lavender	Shankar et al. (2021)
Flavonoids	Lavender, peppermint, grapefruit	Russo (2018)

Continued

TABLE 1.1 Phytochemicals present in different plant-based essential oils—cont'd

Phytochemicals	Essential oils	Citation
Glycosides	Clove oil	Sadiq et al. (2019)
Alkaloids	Mint, orange, rosemary, clove	Misharina and Alinkina (2017)
Phenols	Orange, lemon, mandarin, tangerine, neroli, cinnamon, clove, thyme, oregano, savory, rosemary, neem	Sujatha and Sirisha (2019)
Coumarin	Citrus, Bergamini, bergamot, cinnamon, sweet clove, cassia, lavender, woodruff oil	Ogawa et al. (2020); Silveira et al. (2020)
Saponin	Onion, garlic	Runde (2020); Shahrajabian et al. (2020)

1.3 Natural variation of the oils (chemotaxonomy and environmental effects, etc.)

Generally, the EOs composition varies according to geographical reasons, harvesting seasons in each type of plant. In thymus and *Origanum* species, 1-methyl-4-(1-methylethyl)-1,4-cyclohexadiene (γ -terpinene) and 1-methyl-4-(1-methylethyl)-benzene (*p*-Cymene) are considered as the initial constituents of 5-methyl-2-(1-methylethyl) phenol (thymol) and 2-methyl-5-(1-methylethyl)-phenol (carvacrol). While in the Greek oregano plants, these compounds are approximately considered equal in their potency with stable concentrations even though collected from different geographical regions (with delayed harvesting). This indicates that *p*-cymene forms thymol through γ -terpinene with this theory also supported by *T. vulgaris* both biologically and functionally. Other EOs have exhibited strong antimicrobial properties during or after the flowering season (Hashemi et al., 2017). Other variations have been observed in the EOs obtained from different plant parts. For instance, EO extracted from coriander seeds (*Coriandrum sativum* L.) exhibited different oil composition as compared to the one from cilantro immature leaves (Delaquis et al., 2002; Falleh et al., 2020). Similarly, some EOs are highly volatile and sensitive to thermal conditions incorporated in their extraction and hydrodistillation resulting in chemical alterations in the compositions. This impact can be reduced especially for stem EOs by employing supercritical fluids (carbon dioxide) (Chemat et al., 2020; Fornari et al., 2012).

Seasonal variations have also been observed in some studies along with terpene dynamics variations during the fruit ripening stage. At the semiripe stage, the monoterpenes concentration was high that later decreased. Sesquiterpenes intensively increased during the ripening stage while the ester occurrence was very low. Although the cosmetic industry is growing exponentially, the EOs from fruits and leaves are rarely employed. Moreover, the impacts of environmental and genetic factors are still unknown. Another aspect of the alterations of EOs includes the drying method employed. An investigation explained both dried and fresh ginger increasing the terpene hydrocarbons while decreasing the concentration of gingerol and monoterpene alcohols formation to the complementary acetates. Another study also highlighted that the drying process helped in decreasing the monoterpenes and citrals as compared to sesquiterpenes owing to the significant contrast in volatile conditions among the classes of compounds present (Eslahi et al., 2017; Néri-Numa et al., 2019) (Tables 1.2–1.4).

TABLE 1.2 Characteristic attributes of different extraction methodologies in the processing industry.

Types of extraction	Principle	Advantages	Disadvantages	Examples of plant extraction	Name of plants	Compounds to be extracted	Cosmeticeutics	Sample size	Reference
(SPME) Solid phase micro extraction	Separating the analyte between the sorbent coated fibers and sample	1: Don't require a solvent. 2: Simple. 3: Fast 4: Sensitive	1: Sample must be clean 2: Matrix effect	Rosmarinus officinalis L. leaves of <i>Eucalyptus citriodora</i>		–			Bicchi et al. (2000) , Zini et al. (2001)
(LLE) Liquid-liquid extraction	Dividing the analyte among the two immiscible liquids to extract the compound of interest	1: It is best suitable for thermolabile compounds and azeotropic mixtures	1: Intensive workforce 2: Emulsion formation. 3: Consumption of large number of solvents 4: Difficulty in automate	Leaves of <i>Veronica longifolia</i> kava root		–			Kubátová et al. (2001) , Suomi et al. (2000)
(SPE) Solid phase extraction	Separation of the analyte between the solid and liquid phases for extraction of the target compound	1: Cheap 2: Less organic compound consumption and less time 3: Simple	1: Analyte discoloration. 2: Need filtration 3: Sorbent absorption capacity is limited.	carob fruits <i>Thymbra spicata</i> L. carob fruits		–			Ozel et al. (2003) , Papagiannopoulos et al. (2004)
(DIC) Instant controlled pressure drop extraction	A sudden drop in pressure and high temperature for short time increase the quality of extract	1: Other compounds can be extracted by processing residue 2: Appropriate for essential oil extraction 3: Less consumption of solvent and require less time	1: Efficiency depends on the optimization of the parameters	Cananga oil Orange peel	(Olive) <i>Olea europaea</i> leaves (Grape stalk) <i>Vitis vinifera</i> L. powder	Vanillic acid Oleuropein Luteolin 7 glucoside Ellagic acid Gallic acid Quercetin	Anticancer Antioxidant Antiinflammatory		Mkaouer et al. (2015) , Ribeiro et al. (2015) Georgiev et al. (2014) , Sanchez-Valdepenas et al. (2015)

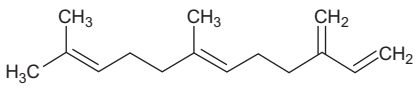
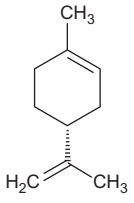
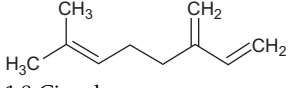
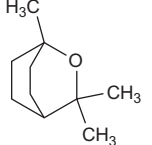
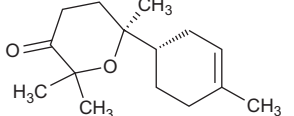
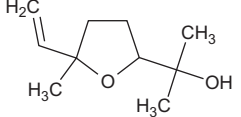
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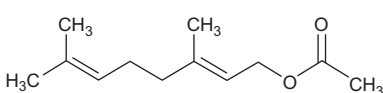
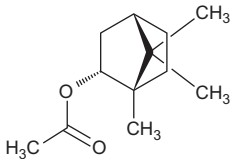
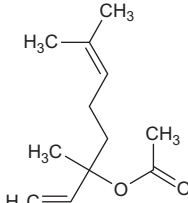
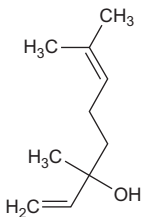
TABLE 1.2 Characteristic attributes of different extraction methodologies in the processing industry—cont'd

Types of extraction	Principle	Advantages	Disadvantages	Examples of plant extraction	Name of plants	Compounds to be extracted	Cosmeticeutics	Sample size	Reference
(UAE) Ultrasound-assisted extraction	The strength of ultrasound allows cavitation which disturbs the cell wall of the sample	1: Less power and solvent consumption, Minimum time 2: Appropriate for the extraction of thermolabile compounds	Different types of plants require different optimization of parameters Cell disruption is high	Lychee seed by-product <i>Prunella vulgaris</i> L. plant	(Haskap berries) <i>Lonicera caerulea</i> L. (Lamiaceae) <i>Thymus serpyllum</i> L.	Anthocyanins Caffeic acid Rosmarinic acid Salvianolic acid	Anti-aging Analgesic agent Anti-septic	600L	Celli et al. (2015) Jovanović et al. (2017)
(MAE) Microwave-assisted extraction	The release of the target compound occurs from the sample when the microwave radiations heats the water	1: Less solvent consumption and reduced time	1: Require filtration 2: Cell disruption	<i>Fucus vesiculosus</i> alga Grape seeds	<i>Myrtus communis</i> L. leaves (Tomato) <i>Solhanum lycopersicum</i>	Gallic acid Galloylquinic acid Naringenin Chlorogenic acid Rutin	Stimulant Hair tonic Anti-allergic	150L	Aleksic and Knezevic (2014) , Dahmoune et al. (2015) Li et al. (2012a, b) , Yamamoto et al. (2004)
(PLE) Pressurized liquid extraction	Hight temperature increases the release of the target compound with the aid of pressure, which enhances filling of extraction cell	1: Less solvent consumption 2: More extraction yield	1: Solvent should be chosen carefully 2: Not suitable for thermolabile compounds	Bran, wheat straw and germ Algae: <i>Himanthalia elongate</i> and <i>synechocystis</i> sp.	(Blackberry) residue <i>Rubus fruticosus</i> L. (Jabuticaba) skin <i>Myrciaria cauliflora</i>	 Anthocyanins	 Anticancer Anti-inflammatory		Machado et al. (2015) , Sariburun et al. (2010) Santos et al. (2013, 2012)
(SFE) Supercritical fluid extraction	Qualities of supercritical fluid use to alter the solvent and selectivity and increase the extraction for the compound of interest	1: Appropriate for compounds having high boiling point and thermally stable 2: Online coupling with the chromatography analysis	1: Need high maintenance	Pitanga leaves Cherry	<i>Coffea arabica</i> (Grape) skin <i>V. vinifera</i>	Oleic acid Palmitic acid Linoleic acid Quercetin Catechin Epicatechin	Antiinflammatory Anti-photoaging Skin conditioning agent	300L	Del Carmen Velazquez Pereda et al. (2009) , Hurtado-Benavides et al. (2016) Belsito et al. (2012) , Chafer et al. (2005)

(SWE) Subcritical water extraction	Extraction process is increased by hot water while the pressure keeps the water in a liquid state	1: Less time consuming 2: Less in cost 3: Environmentally friendly	1: It is selective because of different dielectric constants of the compounds	By product of rice bran biomass By product of onion skin	(Potato) peel <i>Solanum tuberosum</i> <i>Moringa oleifera</i>	Chlorogenic acid Gallic acid Ferulic acid Coumaric acid Quercetin Caffeic acid Quinic acid	Singh and Saldaña (2011) Khoza et al. (2014), Mishra et al. (2011)
(PEF) Pulsed electric fields		Electroporation of the walls cells	Difficult ease of operation				Continuous
Steam Distillation or Hydro Distillation		Large scale production	Temperature limitation				>1000L
Solvent Extraction		Large scale production	Solubility limitation				>1000L
Ohmic heating		Cell disruption is high	Should have known how				Continuous

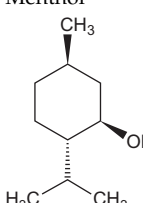
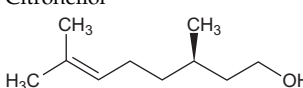
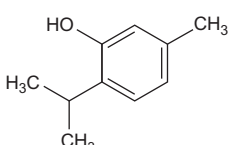
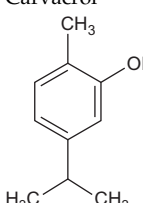
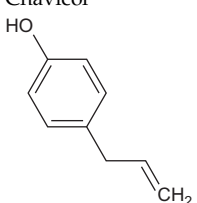
TABLE 1.3 Chemical classes, structural representation, and therapeutic properties of essential oil constituents (Aumeeruddy-Elalfi et al., 2016; Sharifi-Rad et al., 2017).

Compounds chemical class	Characteristics	Chemical structural representation	Therapeutic properties	Examples
Hydrocarbons	They mainly contain hydrogen and carbon molecules. Classified into terpenes (mono-, sesqui-, and di-terpenes) (C10, C15 and C20 respectively). They can be acyclic, alicyclic (mono-, bi-, or tri- cyclic) or aromatic	Farnesene  <chem>CC(=C)CC=CC=CC</chem> Limonen  <chem>CC1=CCC(CC1)C(=C)C</chem> Myrcene  <chem>CC(=C)CC=CC=C</chem>	Antitumor, antiviral, antibacterial stimulant, hepatoprotectant, decongestant	(+)-limonene, (–)-α-pinene, myrcene, sabinene, β-phellandrene, p-cymene, cadinene, fenchane, farnesene, thujane
Oxides	They are strongest odorants with 1,8-cineole being the most widely occurring	1,8-Cineole  <chem>CC1=CC2=C(C1)OCC2(C)C</chem> Bisabolone oxide  <chem>CC1=CC2=C(C1)OCC2(C)C(=O)C</chem> Linalool oxide  <chem>CC1=CC2=C(C1)OCC2(C)C(O)C</chem>	Stimulant, Antiinflammatory, expectorant	Linalool oxide, ascaridole bisabolone oxide and sclareol oxide

Lactones	Lactones reportedly have high molecular weight		Antimicrobial, analgesic, antipyretic, sedative, hypotensive, antiviral	Alantrolactone, bergaptene, psoralen, costuslactone, aesculatine, citroptene, nepetalactone, dihydronepetalactone
Esters	It contributes to a sweet smell, commonly present in many EOs	Geranyl acetate  Bornyl acetate  Linalyl acetate 	Antiinflammatory, antispasmodic, sedative, antifungal, anesthetic	Eugenol acetate, geranyl acetate, bornyl acetate, linalyl acetate
Alcohols	Alcohols have the highest therapeutic contributions without any contraindications	Linalool 	Antimicrobial, antiseptic, balancing, tonifying, spasmolytic, anesthetic, antiinflammatory	Linalool, (+)-citronellol, nerol, geraniol, borneol, menthol, santalol, nerol

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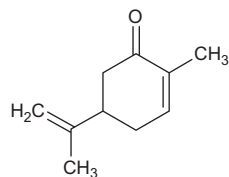
TABLE 1.3 Chemical classes, structural representation, and therapeutic properties of essential oil constituents ([Aumeeruddy-Elalfi et al, 2016](#); [Sharifi-Rad et al., 2017](#))—cont'd

Compounds chemical class	Characteristics	Chemical structural representation	Therapeutic properties	Examples
Phenols	Phenols are considered most reactive, potentially toxic, can cause irritation to the skin and mucous membrane with crystalline structural properties at the room temperature	Menthol 	Spasmolytic, anesthetic, irritant, immune stimulating	Thymol, carvacrol, eugenol, chavicol
		Citronellol 		
		Thymol 		
		Carvacrol 		
		Chavicol 		

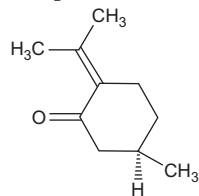
Ketones

Ketones are considered relatively stable and not easily metabolized by the liver. They can also be neurotoxic and abortifacients in the case of thujone and camphor

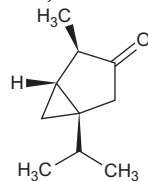
Carvone



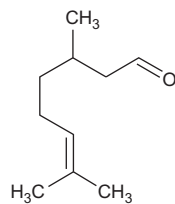
Pulegone



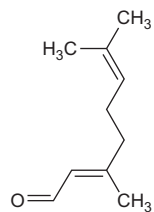
Thujone



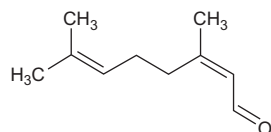
Citronellal



Geranial



Neral



Mucolytic, regenerative, sedative, antiviral, neurotoxic, analgesic, digestive and spasmolytic

Pulegone, menthone, carvone, camphor, fenchone, thujone, verbenone

Aldehydes

Aldehydes irritate the skin and mucous membranes. They exhibit sweet, pleasant fruity smell

Antimicrobial, antipyretic, antiviral, soothing, hypotensive, sedative, spasmolytic, tonic, vasodilators

Benyaldehyde, citronellal, cinnamaldehyde, myrtenal, citral (geranial and neral), citronellal

TABLE 1.4 Summary of LD50 values for some essential oils with the experimented animal and mode of administration (Hashemi et al., 2017).

Plant name	LD50 concentration	Mode of administration	Animals	References
<i>Aegle marmelos</i> L.	23.66 g/kg b.wt.	Oral	Mice	Kuttan and Liju (2017), Hashemi et al. (2017)
<i>Baccharis dracunculifolia</i>	DC >2 g/kg b.wt.	Oral	Rats	Kuttan and Liju (2017), Hashemi et al. (2017)
<i>Cuminum cyminum</i>	>2 g/kg b.wt.	Oral	Rats	Hashemi et al. (2017), Kuttan and Liju (2017)
<i>Eucalyptus globulus</i> Labill	3.8 g/kg b.wt.	Oral	Rats	Kuttan and Liju (2017), Hashemi et al. (2017)
<i>Ligusticum chuanxiong</i>	2.25 g/kg b.wt.	Intraperitoneal	Mice	Kuttan and Liju (2017), Hashemi et al. (2017)
<i>Mentha longifolia</i> L.	470 mg/kg b.wt.	Oral	Rats	Kuttan and Liju (2017), Hashemi et al. (2017)
<i>Vitex negundo</i>	<2 g/kg b.wt.	Dermal toxicity	Rats	Kuttan and Liju (2017), Hashemi et al. (2017)

Season associated with the chemical changes are highly significant in oils extracted from berries as compared to the ones obtained from foliage. Berries oil has reportedly increased by 162% from spring to the autumn season. In summer, the concentration of α -pinene decreased immensely in the foliage oil. This seasonal variation was not as visible in berry oils (Eslahi et al., 2017). Other factors from include edaphic and climate conditions impacting both the quality and quantity was recorded. Furthermore, sunlight exposure contribution is also necessarily considered as a response to the ecological changes. For instance, monoterpenes present in peppermint are used in the breakdown of energy while it degrades itself within the secretory cells on proper storage (Eslahi et al., 2017). Genotype and agronomic factors including age, harvesting time, fertility of the soil, crop diversity also impacted the quantitative distribution of the essential oils in the aromatic plants. Planting time and mineral fertilization also contributed to the overall oil quantity. Similarly, the spring-based oil crops exhibited higher menthol concentration, and reduced terpinene-4-ol concentrations was found in the autumn planted oilseed crops. Further research is needed to understand the difference in the mineral fertilization impacted the oil quantity in plants. The research highlighted that the pulegone concentration increased, while *p*-caryophyllene reduced along with the menthone contents (Hashemi et al., 2017).

1.4 Extraction methods of essential oils

Various parts of aromatic plants are used for extracting essential oils that are later employed for many industries including the cosmetic, pharmaceuticals, and food-based businesses. Different techniques for extraction are adopted owing to the required characteristics

and bioactive components segregation from the extract. Inappropriate extraction methods can result in loss of pharmacological component, stain effect, alteration in the physiology of essential oil, and the off-flavor/odor development in the end-product. Two categories of extraction methods are used: conventional methods and innovative methods. An improvement in the extraction efficiency, isolation time, energy dissipation, yield/output, and quality attributes of EOs is reported with the adoption of innovative extraction techniques (Aziz et al., 2018; El Asbahani et al., 2015; Tongnuanchan and Benjakul, 2014).

1.4.1 Conventional extraction methods

Essential oils extracted by conventional extraction methods utilize the heating process by water distillation.

1.4.1.1 Hydrodistillation

Hydrodistillation (HD), being the oldest and simplest method, was discovered by Avicenna, via alembic. HD equipment constitutes a heating source, vessel (alembic), a vapor condenser device, and a decanter. Initially, the procured source is immersed directly into the water tub present inside the alembic vessel where this mixture is boiled followed by condensation and decantation. Rose was the first plant material used in this HD unit. It is considered unique for the extraction of hydrophobic materials with a high boiling point including wood and flowers. Essential oils are protected by the water without any chance of being overheated. Therefore, a primary advantage of hydrodistillation is its ability to isolate the oil from plant materials below 100°C (Aziz et al., 2018; Tongnuanchan and Benjakul, 2014).

In comparison with Solvent-Free Microwave Extraction (SFME), HD produced a total yield of 0.31% (volatile fraction), while SFME yielded 0.39% in rosemary (*Rosmarinus officinalis* L.) (Okoh et al., 2010). Generally, the HD process has been modified using the current technologies in the ongoing research. Microwave-assisted HD (MAHD) exhibited superiority in both the energy dissipation and the isolation period (75 min) as compared to 4-h of HD (Asghari et al., 2012; Golmakani and Rezaei, 2008). Ohmic-assisted HD (OAHD) is another form of advanced HD discovered by Gavahian et al. In this method, thyme essential oil was extracted in 25 min without any change in the characteristic components of the oil (Gavahian et al., 2012).

1.4.1.2 Steam distillation

It is considered one of the most broadly applied techniques in the extraction industry of oils. Approximately 93% of the EO is extracted via Steam distillation (SD), while the remaining 7% can be attained using other techniques. In this method, the plant sample is heated with steam produced via a steam generator resulting in breakage and release of aromatic constituents and essential oils (Mohamad Kahar et al., 2021). An innovative SD technique was developed by Masango that enhanced the EO yield while reducing wastewater production. It involved placing plant samples (packed bed form) right above the source of steam generation to minimize its loss while avoiding the mixing of botanical samples with water (Božović et al., 2017; Meziane et al., 2020). Similarly, in another study,

2,2-diphenyl-1-picryl hydrazyl (DPPH) was employed to analyze the antioxidant constituents in SD. This led to an increased yield in antioxidants in EO when compared with HD (Aziz et al., 2018; Roohinejad et al., 2017; Yildirim et al., 2004).

1.4.1.3 Hydrodiffusion

Hydrodiffusion employs steam injection to the plant sample container but is limited to predried samples of the plant, otherwise prone to damaging at boiling temperatures. In SD, steam is introduced from the bottom of the steam generator, while in hydrodiffusion, it is incorporated from the generator's top section at lower temperatures or via vacuuming where the steam temperatures can be lowered below 100°C (Aziz et al., 2018; Rezaei et al., 2021). This SD method is upgraded by incorporating microwave technology. A research carried out by Bousbia and the team investigated the difference between innovative Microwave Hydrodiffusion Gravity (MHG), and conventional HD methods. Another study examined the isolation of EO from orange peel using an innovative SD technique (SDf) called microwave steam diffusion (MSDf). Results indicate that the isolation time by MSDf is within 12 min with a similar yield and aromatic profile as compared to SDf with an isolation time of 40 min (Aziz et al., 2018; Calinescu et al., 2017; Ferreira et al., 2020).

1.4.1.4 Solvent extraction

Generally acetone, petroleum ether, hexane, methanol, or ethanol have been incorporated in this solvent extraction technique to extract delicate flower materials otherwise destroyed by steam or heat-based extraction techniques (Tongnuanchan and Benjakul, 2014). In this method, the sample is mixed with solvent by mildly heating it. It is then filtered, and the solvent is later evaporated. The remnants of evaporation process or filtrate entails a resin, or the waxes, aroma, and the EO. This filtrate is combined with alcohol to dissolve EO in it which is later distilled at considerably low temperatures. During distillation process, alcohol after absorbing the aroma will evaporate while the absolute EO stays inside the pot residue. Owing to the complexity of this extraction mechanism, it is expensive and time-consuming in contrast to other conventional extraction techniques (Golmohammadi et al., 2018; Li et al., 2009).

In research conducted on *ptychotisverticillate's* antioxidant profile by solvent extraction, 48% of phenolic compounds were detected along with 44.6% carvacrol and 3.4% thymol (Tomi et al., 2011). Similarly, the EO from *Thymus praecox* subsp. *Skorpilii* var. *Skorpilii* (TPS) were investigated with ethanol, methanol, and water mixture. The results highlight that significant free-radical scavenging activity with 40.31% of thymol and 13.66% of *o*-cymene was present. It was also observed that using water as solvent yielded better phenolics and flavonoids in contrast to other solvent solutions (Ozen et al., 2011; Salehi et al., 2019).

1.4.2 Innovative extraction methods

Extraction methods were modified owing to the hydrolysis, isomerization, and oxidation phenomena observed during some conventional methods. Innovative extraction mechanisms were developed considering the carbon dioxide emission, usage of solvents, energy consumption, and the reduction in extraction time (Aziz et al., 2018; Wang and Liu, 2020).

1.4.2.1 Supercritical fluid extraction

Supercritical fluid extraction (SFE) technique principally uses and recycles fluid in repeated stages of compression followed by decompression. The supercritical state of carbon dioxide (CO_2) is achieved by its heating and implication of high compression. When this fluid form then passes through the plant sample, it loads the volatile matter in it. Later followed by decompression, to segregate the extract from the carbon dioxide. This carbon dioxide is released from 2nd separator and recycles back to the storage tank without any residues in the final product as the carbon dioxide reverts to the gas state under normal atmospheric temperature and pressure (Fornari et al., 2012; Yousefi et al., 2019).

The supercritical state of a fluid is dependent on two factors primarily which are the critical pressure and the temperature, indicated as P_c and T_c respectively. Fluids having these two crucial attributes have low viscosity, high diffusion rate, and density profile similar to the liquids. Therefore, carbon dioxide is used owing to its following characteristics. It reaches critical point with ease at P_c : 72.9 atm and T_c : 31.2°C. It is also considered nonhostile for thermolabile molecules (mainly plants) along with being inert and toxic. It is nonflammable with readily available at a lower cost; is easily eliminated and has high polarity (like pentane) making it suitable for the extrication of lipophilic constituents (Arumugham et al., 2021; Ghannadi et al., 2012).

Comparative studies between supercritical fluid extraction and hydrodistillation method highlight the former's superiority in performance and pharmacological properties. Another study examined the carrot EO obtained by the former technique to exhibit better antibacterial and antifungal properties against *Bacillus cereus* in comparison to the latter one (Glišić et al., 2007; Herrero et al., 2010). Other examples of EO effectively extracted using supercritical fluid extraction include rosa geranium, *Eugenia caryophyllata*, clove buds, and *Marchantia convolute* (Babu and Kaul, 2005; Caniard et al., 2012; Gomes et al., 2007) (Fig. 1.6).

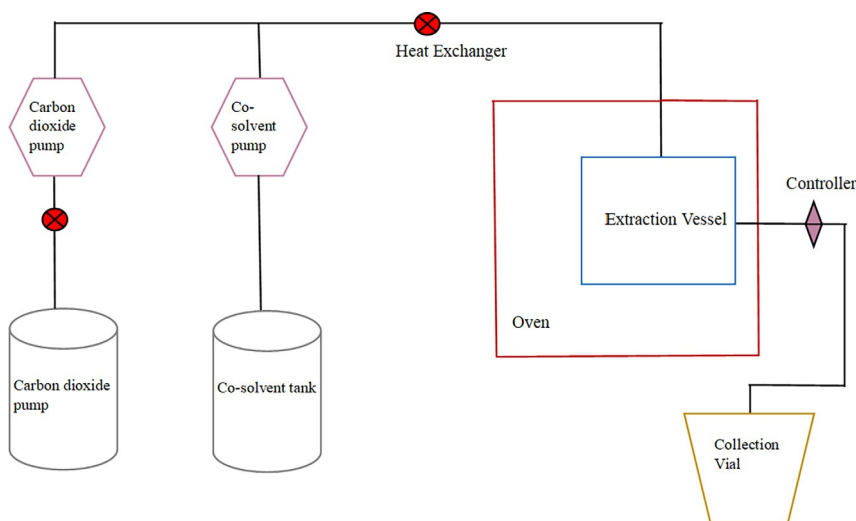


FIG. 1.6 Schematic representation of supercritical fluid extraction.

1.4.2.2 Subcritical extraction liquid

The Subcritical extraction liquid (SEL) technique is considered a very powerful alternate for many EOs. In SEL, the subcritical stage of liquid is when liquid reaches a pressure point higher than the P_c and lower than the T_c , or vice-versa. Water and carbon dioxide are the two fluids used in SEL. Among the superior characteristics contributed by SEL, some include reduced viscosity, lower in density, and an increased diffusion between gas and liquid state. It is also considered quick in the isolation process and can operate at lower working temperatures along with cost-effective extraction and being environmentally friendly (Ibrahim et al., 2019; Nehme et al., 2021; Özel et al., 2006). The comparative analysis indicates that SEL extracted time is 15 min as compared to 3 h of other conventional extraction mechanisms. Similarly, owing to the EOs with significant oxygenated compounds inside without any terpenes concentration can also be accomplished with SEL saving both the energy and plant samples (Aziz et al., 2018).

Research conducted by Kubtova and coworkers compared the lactones extraction from a *Piper methysticum* root using SEL and Soxhlet extraction with water. SEL was conducted at 100°C and 175°C for 20 min and 2 h, respectively. Soxhlet extraction took approximately 6 h for extraction and produced a lower yield by 40%–60% (Kubatova et al., 2001; Nastić et al., 2018; Ni et al., 2021).

1.4.2.3 Solvent-free microwave extraction

SFME technique was developed to overcome the loss of evaporative constituents, poor isolation coherence, toxic residual content in solvents at the final product, etc. SFME is therefore used for expedition of EOs from different spices, herbs, and dried seeds leading to high yield, selectivity, short time of extraction, and environmentally friendly technique (Bousbia et al., 2009; Jugreet et al., 2020). It involves a combination of two existing techniques. Initially, the plant samples are heated using microwave technology which is later passed through dry distillation, operated at atmospheric pressure without any solvent. In comparison with hydrodistillation, the yield of oregano EO was 0.048 mL/g, around 6% lower than the SFME oregano EO yield obtained in another research conducted by Bayramoglu and his coworkers (Bayramoglu et al., 2008; Diniz do Nascimento et al., 2020). Another researcher compared the yield of SFME with the traditional extraction method's yield, environmental impact, solvent residual content, antimicrobial potential, and isolation time. It was reported that microwave extraction offers a shorter isolation period of EO with 30 min as compared to 3 h of hydrodiffusion (HD) and 1 h of cold pressing. Similarly, in terms of yield, SFME exhibited 0.24%, while HD was 0.21% and cold pressing produced 0.05%. Additionally, high energy consumption was observed in HD and cold pressing as compared to microwave extraction and no solvent was used in SFME enabling the process to have cleaner features along with high antimicrobial potential exhibited by the EOs extracted via SFME (Boukhatem et al., 2020; Ferhat et al., 2007; Moradi et al., 2018).

1.5 Characterization of all essential oil

EOs are composed of volatile organic compounds (>300) generally with a lower molecular weight. High vapor pressure of EOs generally, at atmospheric pressure and room temperatures enables them to be partially in the vapor state. Structurally EOs have two major families

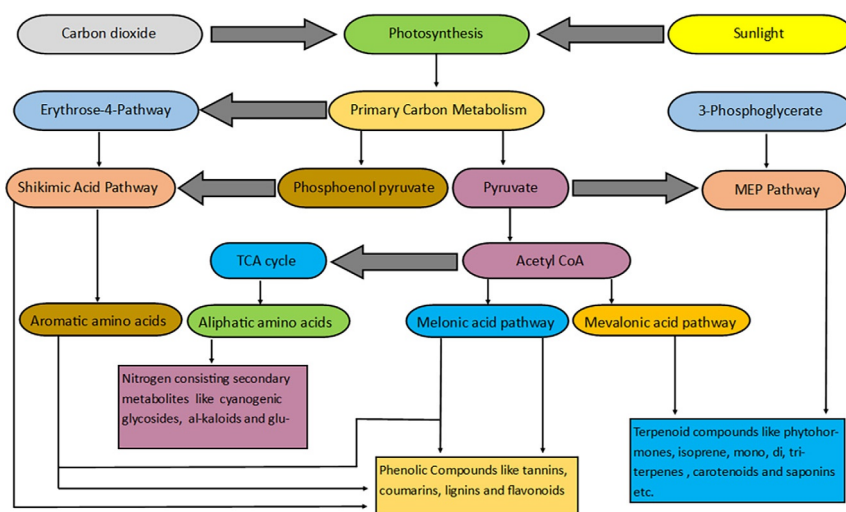


FIG. 1.7 General pathway of biosynthetic pathways of secondary metabolites in plants.

related to their hydrocarbon skeleton, including terpenoids and phenylpropanoids. Terpenoids and phenolic acids have distinct primary metabolic precursors that are biosynthesized through different pathways, as shown in the figure below (Fig. 1.7).

For example, mevalonate and mevalonate-independent pathways are responsible for the biosynthesis of terpenoid, while the shikimate pathway was used for derivation of phenylpropanoid (Sharifi-Rad et al., 2017). Other than phenylpropanoids, terpenes, and terpenoids, numerous other chemical structures are also present contributing to a plethora of properties including antimicrobial, virucidal, insecticidal, antiparasitic, anticancer, antidiabetic, antioxidant, cardiovascular, cosmetic, and food-based contributions (Aumeeruddy-Elalfi et al., 2016). Some of these characteristics are summarized below in the table.

1.6 Applications of all essential oils

1.6.1 Antioxidant

Free radicals and reactive oxygen species (ROS) are associated with inflammation, cancers, cardiovascular diseases, brain dysfunction, immune-related and age-based disorders. Plant EOs are mainly recognized as a useful source of natural antioxidants as depicted in various experimental models (Jugreet et al., 2020). The secondary metabolites and phenolics with conjugated double bonds specifically exhibit the antioxidant properties, for example, monoterpene phenols including thymol and carvacrol, being active ingredients display redox properties. They also perform a fundamental role in the decomposition of the peroxide along with the neutralization of free radicals (Jugreet et al., 2020; Sharifi-Rad et al., 2017). Recent work of EOs as an antioxidant (Elansary et al., 2018; Mehdizadeh and Moghaddam, 2018; Mutlu-Ingok et al., 2020; Valdivieso-Ugarte et al., 2019) include spices (Diniz Do Nascimento et al., 2020), basil (Ahmed et al., 2019), *Thymus vulgaris* and *Thymbra spicata* (Gedikoğlu et al., 2019), *Mentha* spp. (Stringaro et al., 2018), etc.

1.6.2 Antiinflammatory

Many EOs are clinically used for its antiinflammatory potential. They are mainly used against allergies, rheumatism, or arthritis, in a research involving the geranium's cutaneous application on mice, inflammation linked to accumulation of edema and neutrophil accumulation and edema. In research conducted on EO obtained from leaves and flowers of *Cymbopogon nardus* (L.) against egg albumin-induced paw edema model in Wistar rats. Similarly, it also hindered the release of mediators (serotonin and histamine) initially after inflammation (Rungqu et al., 2016). Another study evaluated an in vitro experiment conducted on spearmint, geranium, and lemongrass EOs restraining the human neutrophil's adhering reaction by tumor necrosis factor- α (TNF- α) at lower concentrations. The principal antiinflammatory components include carvone, citral, citronellol, and geraniol; however, in vivo studies have also highlighted the intraperitoneal administration of EOs in mice to be effective in preventing deposition of neutrophil into the peritoneal cavity mainly triggered by the casein injection (Abe et al., 2004, 2003; Mao et al., 2021; Santos et al., 2019).

Moreover, cinnamaldehyde and linalool from *Cinnamomum osmophloeum* Kanei are also found effective in several endotoxin-induced body weight loss and lymphoid organ enlargement in mice in a dose-dependent manner (0.9 and 5.2 mg/kg respectively). Studies have also associated them with the prevention of TLR4 NLRP3 pathway functioning as an antiinflammatory agent (Dorri et al., 2018; Kim et al., 2019; Kumar et al., 2019; Lee et al., 2018).

1.6.3 Antihyperpigmentation

Various dermatological disorders including freckles, melasma, age spots, etc. are caused by tyrosinase enzymes that catalysis the melanin formation in the skin resulting in epidermal hyperpigmentation. The antityrosinase potential of various plant metabolites including EOs has been scrutinized for skin whitening purposes. For instance, *Eucalyptus camaldulensis* Dehnh. Flower EO is used for both antityrosinase and in the reduction of melanin synthesis. EO inhibited the signaling pathway regulating tyrosinase activities along with depletion of ROS in both cellular and intracellular levels (Di Martile et al., 2020; Huang et al., 2015). Similarly, cinnamaldehyde is also known to decline the tyrosinase and melanin content. It is also responsible for the down-regulating the tyrosinase without any cytotoxic display. Likewise, lime mint EO, with β -Caryophyllene constituent mainly decreased melanin in murine B16F10 melanoma cells in a dose-dependent manner. This active ingredient reduced melanogenesis by down-regulating the melanocyte expression including transcription factor (MITF), tyrosinase-related protein 1 and 2, and tyrosinase (Yang et al., 2015). Other antimelanogenic potentials of EOs include *Citrus hystrix* DC, *Citrus reticulate* Blanco, and *Citrus grandis* (IC₅₀: 96 15.92 ± 1.06 , 23.75 ± 4.47 , and 28.99 ± 5.70 $\mu\text{g/mL}$ respectively). This antimelanogenic potential on melanoma cells was evaluated in mice on both intracellular and extracellular levels (Aumeeruddy-Elalfi et al., 2018; El Khoury et al., 2019; Kim et al., 2018; Zhou et al., 2020).

1.6.4 Antidiabetic

There is growing research for natural antidiabetic agents owing to their effective diabetes management with reduced tolerance and side effects. *Pelargonium graveolens* EO is considered effective for hypoglycemic effect along with decreasing lipid peroxidation activity and

improving antioxidant defense mechanism in the body (Al-Sagheer et al., 2018; Boukhris et al., 2012). Meanwhile, *Melissa officinalis* leaves EO significantly reduced the plasma glucose while improving the tolerance of glucose in type-2 diabetic mice. The EO at low concentration was effective for hypoglycemic control, mainly triggered by increased glucose uptake and increased adipose tissues and liver metabolism along with prevention of liver glucogenesis (Chung et al., 2010; Wojtunik-Kulesza et al., 2019).

Menthol, another dietary monoterpene obtained from *Mentha piperita* L. plays a significant role as an antidiabetic agent. The research investigated menthol's role on glucose metabolic enzymes and pancreatic islet cell apoptosis of streptozotocin nicotinamide (STZ-NA) in diabetic rats with 25, 50, and 100 mg/kg menthol and glibenclamide (600 µg/per kilogram body weight basis) for 45 consecutive days. Menthol reportedly decreased the blood glucose levels along with glycosylated hemoglobin while increasing the hemoglobin (total), enhancing the plasma insulin, and escalating the liver glycogen in diabetic rats. Additionally, the hepatic glucose metabolic enzymes and liver damage serum biomarkers were almost also reported to be back to normal levels. Furthermore, other abnormalities in hepatic and pancreatic islets were also diminished. The mechanism adopted was by suppressing the pancreatic β -cells apoptosis, and increase in antiapoptotic Bcl-2 concentration, and decreasing the Bax expressions (pro-apoptosis) resulted in an effective diabetes management (Bouyahya et al., 2020b; Madhuri and Naik, 2017; Muruganathan et al., 2017).

1.6.5 Antimicrobial

The antimicrobial role of EOs has been widely researched with the major drive coming due to the incessant emergence of antibiotic resistance around the globe. The antibacterial capacity of EOs is linked with potential of the membrane being reduced, proton pump interruption, and ATP depletion. These changes can result in a cascade effect impacting the cell organelles. Another research mentioned the changed cell permeability, deteriorated cellular respiration, and increased intracellular escape of K⁺ resulting in alteration of the growth pattern of *E. coli* and *S. aureus* (Aumeeruddy-Elalfi et al., 2016; Mehdizadeh and Moghaddam, 2018). Some bioactive constituents including thymol, menthol, and linalyl acetate also resulted in creating disturbances in lipid fractions, and with altered permeability, the intracellular materials escaped in *E. coli* and *S. aureus*. Gram-negative bacteria exhibit less sensitivity to the EOs as compared to their counterparts owing to cell wall structures (hydrophilic) that restrict the easy movement of hydrophobic compounds via the membrane of the cells (Burt, 2004; Tariq et al., 2019). Similarly, the role of EOs against *Helicobacter pylori*, has been widely researched with its impact against peptic, gastric ulcers, gastric cancers, and lymphomas. Additionally, *Ocimum basilicum* L. and *Salvia officinalis* L. were also researched to decrease the production of biofilm caused by *P. aeruginosa* strains (Percival and Suleman, 2014; Sharifi-Rad et al., 2018; Stojanović-Radić et al., 2016).

Moreover, the antifungal role of EOs of *C. nardus* has been observed against *Candida* species including *C. albicans*, *C. Krusei*, *C. glabrata*, *C. tropicalis*, *Candida parapsilosis*, and *C. orthopsilosis*. These EOs also helped in the disruption and removal of biofilms along with showing antifungal properties better than citronella (De Toledo et al., 2016). Furthermore, it was highlighted those mitochondria and cytoplasmic membrane are the primarily targeted sites for the anti-*Candida* EOs obtained from *Anethum graveolene* L. seeds. Other than

membrane lesions and reduction in ergosterol concentration entering the plasma membrane damage. ROS formation was also observed after the mitochondrial dysfunction, as a key event in the death of *C. albicans*. Other antifungal contributions of thymol EO have been observed against *Fusarium graminearum* where the membrane is damaged by the lipid peroxidation and ergosterol biosynthesis (Gao et al., 2016; Kalagatur et al., 2018).

1.6.6 Antiviral

The potential of EOs against the viruses has also been well researched. The antiviral role of eucalyptus EO, thyme, tea tree (>96%), and their monoterpenes (>80%) have been evaluated in vitro experiments against Herpes Simplex Virus 1 (HSV-1). The EO of *Melaleuca* species has also been virucidal with 99%, 92%, and 91.5% against *M. armillaris*, *M. Leucadendron*, and *M. ercifolia*. The free viral particles were rendered inactive in a dose-dependent way causing HSV-1 inactivation. A combination of different monoterpenes exhibited a 10× higher selectivity index and lower toxicity as compared to individual usage of monoterpenes (Jugreet et al., 2020). Out of 62, 11 EOs contribute as an antiinfluenza virus (A/WS/33) by reducing the cytopathic symptoms (>30%). Anise, marjoram, and clary sage with linalool as a common compound exhibited approximately 52.8% antiinfluenza properties (Choi, 2018). Moreover, the antiherpetic properties of *Thymus bovei* Benth (TBEO) is also evaluated against HSV-2 in Vero cells (having HSV-2 infection) with significant preventive potential observed against HSV-2 replication mechanism. Molecular docking of geraniol, the active ingredient of TBEO highlighted that it acts as a contentious inhibiting agent by binding to the HSV-2 protease, a site used for making the replica of viruses. With acyclovir (ACV), it acts synergistically against HSV-2 highlighting its potency as an HSV-2 infection (Hassan et al., 2018; Trembl et al., 2020).

Another antiviral role was observed by carvacrol active ingredients, extracted from oregano EO against murine norovirus (MNV). It resulted in the losing the infectivity potential of cell culture by impacting the RNA and the capsid of the virus. Carvacrol inactivated MVV within 1 h of exposure along with capsids enlargement from ≤ 35 to ≥ 800 nm in diameter showing promising antiviral potential for treating MNV (Gilling et al., 2014; Ma and Yao, 2020; Sharifi-Rad et al., 2018).

1.6.7 Anticancer

EOs have been tested for suppressing different cancers cell. Among these experimental models, sesquiterpene alcohols specially α -bisabolol (from chamomile and other plants) in EOs are considered as a potent inducer in glioma malignant cells of both the human and rats. Furthermore, the mitochondrial inner transmembrane potential and cytochrome C discharge from the mitochondria of malignant cells highlight apoptotic phenomena. Furthermore, research also indicated that α -bisabolol is nontoxic if the normal glial cell's viability is not affected (Al-Dabbagh et al., 2019; Cavalieri et al., 2004). Camphene extracted from *Piper cernuum* EO also exhibited antitumor properties in melanoma cells. It triggered apoptosis by the initiation of the caspase 3 process and signaling of stress conditions in endoplasmic reticulum (Girola et al., 2015; Rodriguez-Vidal et al., 2020). Moreover, the carvacrol methodology

indicates that phenolic monoterpenes stimulated apoptosis by the mitochondrial permeabilization in the breast cancer cell line of the metastatic origin (MDA-MB-231), resulting in the release of cytochrome C, fragmentation of DNA, and caspase induction via the cleavage of poly ADP ribose polymerase (Aumeeruddy-Elalfi et al., 2016; Quintans et al., 2019).

Additionally, *Thymus bovei* Benth. EO also demonstrated strong cytotoxic activity against the lung adenocarcinoma (A-549-C5), colon cancer (LS-174-D3), and human cervical carcinoma (HeLa-R2) with IC50 values 8.62, 9.30, and 7.22 µg/mL respectively (Hassan et al., 2018). Similarly, different other EO compounds have been used in combination with chemotherapy drugs to improve the cytotoxicity in different cancer cell lines. An example includes D-limonene in combination with the docetaxel sensitized prostate cancer cell line (DU-145) in a dose-depending manner. This combination also suggested a lower dosage of the chemotherapy drug in comparison to the existing usage amount (Jugreet et al., 2020; Sundaram et al., 2013).

1.6.8 Cardioprotective

The therapeutic role of different EOs has been promising against cardiovascular diseases (CVD). On treatment with thymoquinone, active substance of *Nigella sativa* L. a reduction in oxidative stress was observed in the abdominal aorta ischemia-reperfusion injury of the rats. Similarly, cinnamaldehyde also regulated activities as vascular contractile, prevent hypertension in insulin-resistant or deficient animals and inhibit the hypo-responsive attribute among vasodilatory agents (Acetylcholine). It is also reported to avert the hyperresponsivity to the vasoconstrictor agents (KCl/Phenylephrine) in the aortic rings (El-Bassossy et al., 2011; Zhu et al., 2017).

A critical factor in cardiovascular failure and hypertension control is prevention of the angiotensin-converting enzyme (ACE). In this regard, an ACE inhibitor, captopril and geraniol, and EO from *Thymus bovei* were compared. Results indicated 95.4% and 92.2% of geraniol and EO from *Thymus bovei* correspondingly at 15 µg/mL significantly inhibited against ACE as opposed to captopril (99.8% at 15 µg/mL) (Hassan et al., 2018). Molecular docking further indicated the geraniol attachment to the ACE (active cavity) via H⁺ bonding (between the -OH of geraniol and ASP-415). Other cardioprotective examples include 1,8-cineole, with its antihypertensive properties displayed on rats exposed chronically to nicotine for hypertension induction. Oxidative stress and controlling nitric oxide were considered primary reasons for decreased blood pressure (Cai et al., 2020; Moon et al., 2014).

1.6.9 Neuroprotective

A rising concern is reported owing to the severe side effects and dependency development of drugs used against different nervous illnesses. Owing to this, increased appeal in plant and plant-based constituents has also been acknowledged (Jugreet et al., 2020). Antioxidant capacities of EOs have been employed in the treatment of free radicals-induced illnesses, neurological and age-related diseases. An example includes lavender essential oil's application in CNS and stress, anxiety, cerebral ischemia, and depression management. It is also associated

with increased cognitive and nervous system protective effects for Alzheimer due to its antioxidant potential (Xu et al., 2017).

Another study exhibited the neuroprotective potency of *Angelica sinensis* and *Sophora flavescens* EO by increasing the defensive mechanism in cerebral ischemia in experimental rat models. This resulted in enhancing the antioxidant process for lesion pathogenesis by reducing the interleukin 1- β level; TNF- α , and oxidative induced stress. The ischemic injury was recuperated by the brain through the antioxidant homeostasis restoration after reperfusion (Li et al., 2012a, b; Porres-Martínez et al., 2013). Moreover, *Citrus limon* L. peel EO also exhibited a key impact on neurodegenerative conditions including butyrylcholinesterase (BChE) and acetylcholinesterase (AChE) in vitro with dependency on concentration being used. The obstruction of BChE and AChE is associated with the oxidative stress-management induced neurodegeneration potential of EOs (Cutillas et al., 2017; Oboh et al., 2014; Olasehinde et al., 2020; Orhan et al., 2008).

1.6.10 Hepatoprotective

Series of both in vitro and in vivo experiments have been ushered to verify the hepatoprotective capacity of EOs. The EO obtained from *Achillea biebersteinii* Afan was evaluated against the hepatotoxicity induced by CCl₄ in rats using various markers. A significant decrease was observed in the increased hepatotoxicity-linked parameters with 0.2 mL/kg EO administration. These results synced with histopathological analysis. The hepatocytes from the counter section exhibited extreme necrosis and deterioration in hepatic lobules. Moreover, the hepatic malondialdehyde (MDA) obtained from lipid peroxidation in the nontreated rats was very high while the group pretreated with EO exhibited a clear decrease in the MDA proportions at the final stage (Al-Said et al., 2016; Daoudi and Bnouham, 2020). Another study involving the prior treatment of rats with EOs (50 mg/kg for 15 days) from *Salvia officinalis* and *Thymus capitatus* were exposed to paracetamol intoxication; indicated that EOs sheltered hepatic lactate dehydrogenase (LDH) and other serum pursuits. An increase of SOD and GPx in the blood and liver was also observed. The hepatoprotective role of EOs was associated with their antioxidation capacity (Banna et al., 2013; Bouyahya et al., 2020a).

1.7 Safety concerns

The history of EO application in preservation, as a food additive, and in traditional toxicology has established the safety of their usage. Some are mixed with different volume of diluents that can result in the intoxication of the body or the skin. Different results on animals indicate the LD50 range of EOs from 1 to 20 g/kg body weight with few deviations. In the cosmetic industry, numerous applications in antiseptics, liniments, soaps, deodorant, flavors, and cosmetic ingredients are observed. Tea tree EO is used in devising hand washing liquids, antiseptics, and mosquito repellents in many countries owing to the monoterpenes and sesquiterpenes concentration present. Similarly, many EOs are employed widely in the mosquito repellent preparation, perfume and fragrance industry (Hashemi et al., 2017).

In agriculture, they are also used as natural pesticides without imparting toxicities. Some insecticidal EOs have also been researched to impede arthropodic neural network via GABA, octopamine synapsis, and the acetylcholinesterase inhibition (Isman, 2020; Regnault-Roger et al., 2012). In aromatherapy, they are absorbed via olfactory mucosa stimulating the limbic system. On interaction with the neuropsychological network, physiological and psychological effects are apparent. This therapy is associated with a positive contribution in labor, infections, dementia, and anxiety treatments; however, it still needs scientific validation. EO from rosemary is associated with hyperexcitability on cell stage owing to the loss of sodium or potassium gradient.

Wound healing time reduction is also associated with the application of tea tree oil while lavender EO is linked to reduction in anxiety presurgery (Hashemi et al., 2017). Similarly, a cutaneous irritation study in human patients was also conducted with *Clausena dentata* resulting in no harmful impact on the human skin but with few side effects (De Souza et al., 2020; Rajkumar and Jebanesan, 2010). The eye safety aspect of EOs includes the usage of 5% tea tree oil for eyelid scrub treatment for nullifying Demodex infestation with improvement in subjective ocular symptoms. Another study supported the usage of tea tree essential oil for the reduction in eyelid scrub, Demodex infection in eyelids, and corneal pathological conditions along with conjunctival inflammation (Gao et al., 2012; Koo et al., 2012). Ophthalmic observation also highlighted that turmeric EO in rats at 1 g/kg consumed orally did not cause any constriction or retinal vessels branching in any treated groups. Therefore, oral administration of turmeric essential oil was ruled out for causing glaucoma and intraocular changes resulting in inflammation (Liju et al., 2013; Soleimani et al., 2018).

Phototoxicity is chemically induced skin irritation due to light. Some EOs from lemon include furocoumarin resulting in causing phototoxicity. Compounds responsible for phototoxicity are identified as oxypeucedanin and bergapten. EO extracted from *Tagetes minuta* and *patula* are also phototoxic therefore not used in the preparation of sunscreen products.

Moreover, some EOs can be carcinogenic if they are activated after metabolism. These if stimulated may induce the malignancy dependent on estrogen production. Usually, this antimutagenicity and mutagenicity are tested on bacteria, mammalian cells, and insects where the studies concluded that EOs can damage mitochondrial dysfunction and can result in respiratory deficient directly or indirectly (Hashemi et al., 2017). Multipurpose benefits of EOs have bagged acceptance globally but the toxicological studies have revealed that some constituents need to be evaluated carefully before legally marketing them for food or drugs. Therefore, in vivo experiments are needed to evaluate their toxicological status to enable further therapeutic, scientific, economic, and conventional applications (Hashemi et al., 2017).

1.8 Conclusion

Among the numerous methods employed for the extraction of EOs, nonconventional ones have been evaluated for their recovery, short extraction time, and higher efficiency, especially ultrasound microwave, supercritical fluid, etc. Other advantages include the reduction in

energy, wastewater, and solvent cost. However, further studies and experiments are needed to analyze the extraction efficiency of hydrophobic and hydrophilic bioactive agents. With various unique and complex chemical compositions, the EO with a biochemical mechanism of action against pathogen is specific against each pathogen. Permeability of the bio-membrane, polarity, cytotoxicity, mitochondrial and genomic toxicity, or electrolyte loss are some likely causes for carcinogenic and antimicrobial effects. Further clinical research is needed to identify the dosage for triggering single or synergistic effects of EOs in food industries, or holistic medicines without imparting any dangers of triggering side-effects (both humans and animals).

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